

Ahmed Mahmoud

Penetration Tester

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Summary

Cybersecurity student with hands-on experience in web, API, and network penetration testing, vulnerability assessment, and digital forensics. Skilled in network traffic analysis, ML-based threat detection, anomaly detection, and AI-powered security solutions. Actively contributes to the security community through bug bounty research and technical writing.

WORK EXPERIENCE

Information Security Intern 10/2025 – 03/2026
CyberThos Remote

Worked on SOC operations, penetration testing, and digital forensics. Performed log analysis, network traffic analysis, and vulnerability assessments using Splunk, Wireshark, and Metasploit.

Machine Learning Intern 11/2025 – 06/2026
Digital Egypt Pioneers Initiative - DEPI Cairo, Egypt

Built and trained machine learning and deep learning models using Python, applying anomaly detection techniques and developing AI solutions with Azure AI, MLflow, and Hugging Face.

Cyber Security Intern 08/2025 – 09/2025
NTI Smart Village, Egypt

Gained hands-on experience in vulnerability assessment, web and cloud security, and threat detection using Nessus, Burp Suite, Splunk, ELK Stack, ScoutSuite, and Prisma Cloud.

PROJECTS

OT Sentinel AI – ICS/OT Detection & Response

Developed an AI-powered ICS/OT detection and response platform using XGBoost and Autoencoder, implementing the AI detection pipeline with feature engineering, risk scoring, and FastAPI integration, validated on GRFICSv3 and Modbus TCP traffic.

Network Traffic Forensics for Intrusion Detection

Developed a real-time IDS using Suricata, Wireshark, and XGBoost for malicious traffic detection and network monitoring.

Botnet Detection System

Furssah AI Competition (CrossRealms ME)

Developed an ML-based botnet detection system for real-time malicious network traffic classification.

IoT Wi-Fi Honeypot

Developed an IoT Wi-Fi honeypot to monitor device behavior during Evil Twin attacks.

AI-Powered Deepfake & Image Manipulation Detection System

Built a digital forensics system for detecting AI-generated, spliced, and manipulated images using machine learning.

Custom Firewall System

Developed a Linux-based firewall using iptables and SQLite for persistent rule management.

SKILLS

Offensive Security

Web & API Penetration Testing, Bug Bounty Hunting, Burp Suite, OWASP ZAP, Nuclei, ffuf, Reconnaissance, Vulnerability Assessment

Network & Cloud Security

TCP/IP, DNS, HTTP/HTTPS, Firewalls, Linux, Windows Server, Active Directory, Network Security, AWS Security Concepts, IAM

ML for Threat Detection

Anomaly Detection, Network Traffic Analysis, Feature Engineering, XGBoost, Autoencoder, TensorFlow, Scikit-learn, Model Evaluation

Soft Skills

Technical Documentation, Team Collaboration, Continuous Learning

SIEM & Monitoring

Splunk (Search, Dashboards, Alert Investigation), ELK Stack, SIEM, IDS/IPS, Log Analysis

Incident Response

Alert Triage, Threat Hunting, Incident Handling, Escalation, Documentation

Programming

Python, Bash, JavaScript, PHP, Flask, Java, SQL

Professional Development

eJPT / eWPT / eMAPT — Self-studied (exam not yet taken)

EDUCATION

B.Sc. in Network and Cyber Security

2023 – 2027

Elsewedy University of Technology - Polytechnic of Egypt

Languages

English — Professional Working Proficiency Arabic — Native

Interests

Bug Bounty Hunting | Security Research | Open-Source Security Tool Development | Technical Writing